Supplement: methods, sensitivities and audit trail

For *Older than their fossils? What the Ordovician–Silurian record can test about the maxima that divide land-plant molecular clocks*.

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Internal research draft from a living repository. Release and submission are Barry W. Brook's decision.

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This supplement holds the machinery behind the main text. Repository paths are given so that every number can be traced to the table or script that produced it. Internal labels appear here because the repository uses them: **M1**, **M2** and **M3** are the 515.5, 459.1 and 431.2 Ma maxima; **E1** trilete spores, **E2** cryptospores accepted as embryophytic under a named reading, **E3** polysporangiate bodies, and **E5** crown vascular-plant members. **B0**, **B1** and **B2** are the body-record categories: any land-plant body (any accepted macro- or mesofossil; canonically including the Katian Oman sporangial fragments, S5.1), polysporangiophyte body and crown vascular-plant member. `manuscript/estimands.md` states what each analysis bounds and which phrasings are not allowed.

S1. Readings, observations and units

Named readings. Disputed identifications, affinities and ages are carried as alternatives, not resolved by fiat. The *main* reading accepts what the cited sources accept. The *inclusive* reading adds records that a named source accepts against others, and the *outlier* reading admits further records that are explicitly contested. A wider reading requires a record-specific acceptance; it does not promote every unresolved grain or axis to a detection. The E2 rule separates spore morphology and producer evidence from the embryophyte affinity accepted by each reading (`data/units/e2-unit-rule.md`). Rejected, reassigned and disputed records are listed in `docs/rejected-records.md`.

Observations. Within each frame, the same reporting rule applies to older and reference material. An explicit non-detection is scoped to the material actually examined. A missing report becomes a zero only after a full-category reporting audit. Unresolved identification, missing links and unreported effort remain distinct from absence.

Ages. Stage bases follow ICS v2024/12 (`data/timescale.csv`). Source-specific age statements and their numerical conversions are retained separately. Literature before 2024 mostly uses GTS2012 Early Devonian boundaries, which differ from ours by up to 3 Myr. Uniform within-range ages and shared within-range positions are modelling choices, not measured dating distributions.

S2. The Murzuq encounter comparison

S2.1 Source and frame

Abuhmida (2013) reports palynology from six Murzuq Basin wells. All 164 appendix samples were transcribed independently, with 158 chart depth and type labels (153 exact matches). The 164 prose-scope rows and 158 chart reports were mapped, retaining barren material and possible caving (`data/units/libya_*.csv`; `docs/audits/2026-10-31-libya-event-map.md`). The primary frame is the chart-coded cuttings design `LY-MZ-CU`. Five wells contribute 59 Llandovery reference cuttings (51 with trilete reports), and three of them contribute 28 older cuttings without trilete reports.

S2.2 Model and qualification

For an eligible unit, the detection probability conditional on a well effect is $1 - \exp(-rux)$. Here r is the reference rate, $u \sim \text{Gamma}(2, 2)$ is shared within a well, and $x = \rho f$ is the older relative intensity. The older/reference delivery-and-recovery ratio ρ takes the declared values 1, 0.5, 0.25 and 0.1; it is not estimated from the source. Reference and older outcomes share the well effect and are fitted jointly.

H0c constrains constant intensity; H0 and H0m constrain exposure-weighted intensity under their declared history classes; H1 allows the declared higher ceiling. The origin read-out at 459.1 Ma concerns the interval from the reference boundary $\tau = 443.1$ Ma to that age. A window comparison instead tests a persistent minimum intensity on an older interval beginning at 459.1 Ma, allowing zero intensity between that boundary and τ. An unbounded low-intensity window therefore does not show that the same intensity is compatible all the way to the reference.

Inference is by test inversion with a declared, simulation-calibrated critical value (`docs/f3-shape.md`). The potential screen, eligibility and estimation path were simulated on a declared finite grid. Results are nominal 95% compatibility bounds, qualified per quantity and design against a 0.93 coverage floor (calibration target 0.955), conditional on passing the potential screen, with abstentions counted as covering. The largest observed marginal miss proportion was 38/1,000. At the B4 boundary of the 462.6 Ma window, 20 of 1,000 screen-passing simulations excluded the truth, and 60 delivered non-abstaining window claims, so 20 of those 60 missed. Population 95% coverage, coverage conditional on a non-abstaining result, simultaneous coverage and validity beyond the tested histories are not established. Thirteen Murzuq design/delivery combinations qualify. A fourteenth qualified pair, QB-CO, supplies no older M2 interval and has a narrower class scope. The receipts are in `models/f3/out/qual/`.

S2.3 Contamination and an identification limit

Murzuq's palynofacies fractions use a different preparation and a 200-particle denominator. They cannot supply separate trilete and cryptospore counts for the chart search, so binary observations are retained and contamination enters as a sensitivity model. Let g be the indigenous fraction of reference *encounter intensity* (not of positive samples), and κ the transfer of the contaminant component into older cuttings. For age-exposure weight q_i, an older zero has probability

$$\exp[-ru\kappa(1-g)]\{(1-q_i)+q_i\exp[-ru\rho gf]\}.$$

The likelihood depends on these parameters only through $a=\rho gf$ and $b=\kappa(1-g)$, so tuples preserving both are observationally equivalent within the model. If every $q_i=1$, only their sum enters the older-zero likelihood. Declining indigenous output can thus be offset by delivery and reference contamination along stated parameter combinations. This is an identification result within the declared model, not a universal property of binary observations.

A predeclared 5,760-row factorial grid varies delivery, indigenous fraction, transfer, allocation, chronology, interval and intensity. Independent uniform ages are compared with fixed, depth-ordered 10%, 50% and 90% positions within source ranges. The threshold $-2\log(0.05)$ is an uncalibrated comparison convention. A contamination scenario that conflicts with older zeros even at $f=0$ is flagged separately. The 384 break-even entries fix $\kappa=b=0$. These surfaces and the 36-case well/age influence exercise are unqualified sensitivities.

Table S1. Murzuq sensitivities (`models/f3/loop2/out/`; `docs/loop2-sprint1-results.md`).

Challenge at $\rho=1$	Observed sensitivity	Scope
Five leave-one-well-out cases	H0 0.1850–0.3646; window endpoints 462.6–468.1 Ma	one factor at a time; unqualified
All nine influence/age/allocation cases	H0 0.1775–0.3646	unqualified
$\kappa=0$, independent uniform stage ages	critical g: 0.0910 for constant origin intensity; 0.9559, 0.2363, 0.1735 for windows ending 462.6, 467.1, 469.4 Ma	$f=1$; comparison contours, not confidence limits
Shared midpoint ages	no exposure in the 459.1–462.6 Ma window	fixed chronology scenario
Shared 90% ages	no exposure in the window ending at 467.1 Ma	fixed chronology scenario

S2.4 The calibration material: Borenshult-1

Hooper et al. (2026) base the 459.1 Ma maximum on the oldest possible trilete spores from the Borenshult-1 core, Östergötland, Sweden (Rubinstein and Vajda 2019). Their Supplementary Table 4 keys the soft maximum to *Chelinospora?* sp. at 48.2 m (*Baltoniodus variabilis* zone, lower Sandbian) and takes the value from the GTS2012 Sandbian base plus its uncertainty, 458.4 + 0.7 = 459.1 Ma. The same rule on our timescale gives 458.9 Ma. They did not use the Borenshult spores for a minimum because the trilete marks might be compression artefacts.

Rubinstein and Vajda's Fig. 3 separates definite from possible triletes. It plots definite triletes at six depths from 43.1 to 33.05 m, all Sandbian or younger and the lowest below the 454.41 Ma Kinnekulle K-bentonite, and possible triletes only at 48.2 and 34.0 m. Their Fig. 4 lists no trilete taxon at 41.1 m, so five depths are concordant between the figures. We carry three readings: Fig. 3 definite, Fig. 4 definite and the concordant five, each with the two possible-only depths unresolved, plus Hooper et al.'s wholesale exclusion as a sensitivity (`data/units/borenshult_review/`, `docs/audits/2026-11-05-trilete-evidence-rubric.md`). Symbols and positive depths are not specimen counts.

The readings bear on three separate questions. (1) The maximum: the stage-base rule gives 459.1 Ma from the definite 43.1 m level as from the possible 48.2 m one. (2) A minimum: the definite triletes would set a crown-embryophyte minimum only under an added mapping of their producers to the crown; that mapping is not supplied by the mark, and its minimum would be the young edge of the bentonite age, not the level itself. (3) An absence test: only three Darriwilian samples lie below the first definite trilete. With all three fully exposed, the screen statistic at reference intensity is 1.3157, 1.0753 and 1.1155 for the three readings, against 5.9915, so the core cannot exclude reference-level intensity before its first definite trilete under any

reading (`models/f3/out/borenshult_review/screen.csv`). This screen has no numerical chronology and is not a test at 459.1 Ma.

The comparison of the earliest trilete claims on common axes (specimen basis, the describing authors' confidence, independent age control, artefact evidence and documentary consistency) keeps Qusaiba-1 as the secure comparator, retains the Turkish occurrence with its original Ashgillian–Hirnantian age wording, and keeps the Hirnantian Zbrza marks outside strict trilete occurrences because their authors doubt them. No composite score is computed.

S3. QSIM-801: a model-conditional floor calculation

Vecoli et al. (2017, Table 1, p. 321) report 99 cryptospore morphotaxon occurrences across 17 of 20 tabled samples assigned to the late Dapingian–early Darriwilian. Their p. 330 explicitly reports no triletes in the studied sequence. The table was independently reconstructed from the rendered source (`data/units/loop2/replication/`). Its 22 categories include informal morphotypes. Marks are distinct recorded occurrences, not counts of independent producers or dispersal events. The table-defined depths are 675.1–767.8 ft. The methods' shallower endpoint, 586.9 ft, is inconsistent and is not used.

Let θ_i be the trilete/control unit-intensity ratio in sample i, and $h > 0$ a relative effective recovery factor. Suppose the reported objects form a co-ascertained, independent target/control sample. Conditioning Poisson counts on the true land-spore total N_i then gives target probability $h\theta_i/(1 + h\theta_i)$. For a persistent floor $\psi = \min_i \theta_i \geq \theta$, the all-zero probability is at most $(1 + h\theta)^{-\sum_i N_i}$. If each mark contributes a distinct target-detectable unit, $M \leq \sum_i N_i$ permits a conservative rejection rule: no targets and $(1 + h\theta)^{-M} \leq \alpha$. Randomness of M does not invalidate the rule: conditional on the actual totals, rejection is a subset of the all-zero event with enough units to cross the threshold. Then $U = \{\alpha^{-1/M} - 1\}/h$ bounds the *minimum* ratio, not every sample's ratio, a mean, or the pooled share.

Table S2. Model-conditional floor limits; none is an admitted empirical confidence bound. D divides the exponent as a heuristic for clustering, not a validated correction.

Relative effective recovery h	Independent units, $D = 1$	Heuristic $D = 2$	Heuristic $D = 4$
1	0.031	0.062	0.129
0.5	0.061	0.125	0.257
0.25	0.123	0.250	0.515
0.1	0.307	0.624	1.287

The source specifies preparation but not the search coverage needed to show that every mark represents a unit on which triletes were detectable. Independence, effective ascertainment and sampling and stopping conditions are unverified. The calculation therefore stays exploratory. Toy Poisson simulations check the algebra but cannot validate the historical protocol. No matched counted Katian–Llandovery reference was recovered. Qusaiba-1 supplies reference context only. Röstånga and Caspalá/Lipeón are younger than M2, and Hanadir/TAYM-4 is separate material without a qualified design (`manuscript/external_candidates.csv`).

S4. The 515.5 Ma maximum (M1): affinity and the preservation-control argument

Morris et al. (2018b) place the maximum at the base of the *Olenellus* Biozone, about 515.5 Ma, below the oldest possible total-group records (Stage 4 Rome Formation cryptospores), and argue for a real absence before it: land-plant spores share the preservation potential of the marine algal cysts with which they usually occur, so older cyst-bearing sequences without land-plant spores record absence (Morris et al. 2018a). Our earlier drafts described only the placement (`docs/audits/2026-11-07-m1-rationale.md`). The control shows that sporopollenin-walled microfossils survived and were recovered; it does not by itself show comparable terrestrial delivery, coverage of the land areas where early plants grew, sensitivity of preparation and search to small cryptospores, or production of recognisable spores by the earliest producers. The assessment below concerns whether the assembled material can quantify the argument.

No assessed frame admits an E2 older-zero/reference design. The Borenshult 71.1 m zero is secure under the main reading but belongs to a frame that fails its potential screen. It is not a pre-515.5 Ma observation. Strother (2016) places the Tennessee taxa in Charophyta. Taylor and Strother (2024) interpret the packets as pre-embryophytic, and K. Wang, Xu, and Yin (2022) favour ancestral aeroterrestrial algae while calling direct

embryophyte affinity unlikely and the interpretation premature. None explicitly accepts the assessed Stage 4 producers as embryophytes. Wall ultrastructure and developmental features bear on precursor relationships, so the inclusive reading stays an explicit alternative. Argentine sample RIOCAP-7999 is unresolved under main and detected under inclusive and outlier readings. Stage 4 converts to 514.5–506.5 Ma, all younger than 515.5 Ma by the maximum's design (`data/units/loop2/m1-m3/`).

S5. The positive body-record analysis

S5.1 Inventory and units

Names. The 241 distinct PBDB macrofossil names overlapping 445.2–393.47 Ma were classified by `gpt-6.1-sol` with web search. Each name received a category, polysporangiophyte and crown-member flags, a basis, a cited source and a confidence (`data/macro/name_classifications.csv`). Names judged implausible for the window, for example the Cretaceous angiosperm leaf genus *Protophyllum*, cannot enter any category. A stratified source audit is in `docs/audits/2026-11-01-loop3-source-review.md`; it does not certify all 241 assignments.

Material identity. Only explicit collection or occurrence identifiers justify replacing a PBDB record with its curated interpretation. Country, taxon and overlapping ages do not establish identity; the pilot used them and falsely merged distinct localities (S7). Regional checklists and unresolved duplicates are withheld from the primary count and kept in a sensitivity. Source decisions are in `data/macro/source_adjudications.csv`. Example: the two Yea horizons are kept as separate units with the full disputed Gorstian–Emsian range, because the Gorstian attribution rests on a disputed graptolite identification. Neither a fixed Gorstian nor a fixed Devonian age is justified.

Units. The primary unit is a locality–age proxy: PBDB formation plus coordinates rounded to 0.1° and identical age support, or a curated named site plus identical age support. For a collapsed unit, the oldest-member age has support from the maximum of member lower ages to the maximum of member upper ages. Missing geographic keys retain the collection identifier. The proxy is conservative but is not a verified bed (`models/f3/loop3/out/primary_unit_membership.csv` lists every constituent record).

Older screen. A separate screen covers all 556 older-only non-spore PBDB rows (69 names). Source review of the seven plausible or unplaced names (17 occurrences) found cyanobacteria, algae or trace fossils and the rejected *Boiophyton*, not an additional embryophyte body. These exclusions are not fossil non-detections, and the screen is not a search denominator (`data/macro/older_records_screen.csv`; `docs/audits/2026-11-01-loop3-older-records-review.md`).

The Oman sporangial fragments. B0 was declared as any accepted land-plant macro- or mesofossil, sterile axes included (`docs/plan-loop3.md`). The inventory as first built took curated records from the E3–E5 channels only, so the spore-filled sporangial fragments of the Ghaba-1 borehole, Oman (Wellman, Osterloff, and Mohiuddin 2003), recorded in E2 (observation S-w22-K12), never entered it. The primary report describes fragments from three samples of one core, some with a sporangial covering around in situ cryptospores; it dates them Caradoc, and the later review we follow correlates them to the Katian. They lack axes and conducting tissue, so they stay out of B1 and B2, but the declared B0 admits them. The omission was an inventory-assembly error, so the Oman-inclusive B0 is canonical and the former inventory is reported as an exclusion sensitivity. The fragments enter as one locality–age unit (Katian enclosure 452.8–445.2 Ma), not as twelve fragments or three samples. Codex's membership audit of all 111 E2 observations found no other record that passes the declared B0 before the Silurian (`docs/audits/2026-11-07-b0-membership-review.md`); the older Hawaz sheets lack secure land-plant tissue identity. Their recovery through palynological preparation differs from the macrofossil collections that dominate the recording proxies, which is a further reason the B0 histories may misdescribe the sequence. The B0 values here come from `models/f3/loop3/oman_sensitivity.py`, which first reproduces the 48 frozen B0 output cells (two readings, four cutoffs, six histories) to floating-point tolerance and then adds the unit.

Reading	Oman	Units	Oldest	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
main	excluded	75/77	432.9	434.51	436.38	435.73	437.16	none	none
main	included	76/78	452.8	455.19	456.62	456.09	465.21	none	none
inclusive	excluded	79/81	445.2	447.20	448.27	447.85	452.98	none	none
inclusive	included	80/82	452.8	455.06	456.43	455.92	464.40	none	none

Table S3b. Land-plant body (B0) bounds (Ma) with the Katian Oman fragments included (canonical) and excluded, Emsian cutoff. Units: guaranteed/possible; Oldest: oldest possible age. History abbreviations:

Const., constant; PBDB, proportional to PBDB land and marginal-marine collection density; PBDB −plant, the same with plant-matched collections removed; Decl. 25 and Decl. 10, exponential decline into the past on 25 and 10 Myr timescales; −plant × decl. 10, their product. "None": no bound within the 469.4 Ma limit of the declared domain.

With the fragments, the oldest unit is Katian, followed by more than 12 Myr with no body unit, and every declared history fits the sequence poorly (Table S7). The bounds move about 20 Myr older. Under the inclusive reading without Oman, the Hirnantian Zbrza axes are the oldest unit and give 447.20 Ma under constant intensity. Neither inventory supplies a lag or delay: the 458.2 Ma cryptospore value is a fossil minimum, not an origin, and the bounds are model-conditioned endpoints. The comparison shows how far an endpoint bound depends on its oldest admitted unit.

S5.2 The endpoint calculation

For a fixed younger cutoff c, let $U(a) = \int_c^a w(s)\,ds$ be cumulative recovery. Conditional on n independent units with ages drawn with density proportional to w on $[c, T]$, the maximum A satisfies $P(A \le a) = \{U(a)/U(T)\}^n$. Inverting at 0.05 gives the exact one-sided 95% older bound

$$U(T_{\text{upper}}) = U(A)\,0.05^{-1/n}.$$

The absolute recovery rate cancels. For constant w this is the Strauss and Sadler (1989) and Marshall (1990) expression with a fixed cutoff, which uses all n observations. Marshall (1994, 1997) and S. C. Wang et al. (2016) relax the assumption of uniform recovery in other ways. A bound beyond the declared history's domain (469.4 Ma, where the held PBDB effort series ends) is reported as *not bounded within the domain*, not as an infinite-age inference.

Age envelope. The primary age treatment uses the greatest possible oldest age and the smallest number of units guaranteed older than c. The bound increases with A and decreases with n, so this is an envelope of the exact-age bounds whenever every reported age interval contains the true age. That geological condition is separate from statistical coverage. Independent-uniform and shared-quantile allocations (256 draws, seed 20261102) are reported as sensitivities. Their quantiles are not confidence bounds (Table S8).

Histories. Six histories are fixed without fitting them to the focal fossils:

1. constant recovery in calendar time;
2. PBDB all-taxa land and marginal-marine collections per Myr, apportioned by age overlap;
3. the same, with collections matching the held plant-filtered extract removed;
4. exponential decline into the past, 25 Myr timescale;
5. exponential decline into the past, 10 Myr timescale;
6. the plant-excluded density multiplied by the 10 Myr decline.

Plant exclusion is a response-related filter, not verified independent effort. No proxy establishes actual search opportunities. Cutoffs are the ends of the Emsian, Pragian, Lochkovian and Pridoli.

Shape diagnostics. Conditional on the maximum, the remaining unit ages should be distributed as $U(a)/U(A)$. Kolmogorov–Smirnov statistics at midpoint ages and across age draws expose histories that fit the observed sequence poorly (Table S7). Table S7 reports every declared age treatment: midpoint ages, independent uniform draws within each range and a shared quantile applied to all ranges. No probability law for ages within ranges was declared, so none is preferred. For crown vascular-plant bodies, constant recovery fits under independent draws but not at midpoint ages or shared quantiles; the 10 Myr decline fits poorly under all three. The robust result is discrimination against steep declines, not support for constant recovery. The p values are asymptotic Kolmogorov–Smirnov values computed on allocated ages treated as exact, not exact finite-sample tests, and their spread across draws is a set of percentiles, not a combined test for interval-dated fossils, a model-selection criterion or an empirical calibration. Because a history that fails under the examined allocations may fit another allocation the intervals admit, an independent review constructed admissible allocations under which the 25 Myr decline is not rejected for the 50 possible crown-member units; an interval-aware analysis is in hand, and until it is complete only the 10 Myr decline is treated as disfavoured.

S5.3 Results and sensitivities

Reading	Category	Units	Oldest	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
main	Land-plant body, without Oman fragments	75/77	432.9	434.51	436.38	435.73	437.16	none	none
main	Polysporangiophyte body	70/72	432.9	434.62	436.63	435.94	437.50	none	none
main	Crown vascular-plant body	48/50	426.7	428.84	428.88	428.45	431.63	none	none
inclusive	Land-plant body, without Oman fragments	79/81	445.2	447.20	448.27	447.85	452.98	none	none
inclusive	Polysporangiophyte body	74/76	445.2	447.34	448.48	448.03	453.62	none	none
inclusive	Crown vascular-plant body	53/55	426.7	428.63	428.67	428.28	431.10	none	468.65

Table S3. Conservative 95% older bounds (Ma) by category, reading and recording history from the frozen inventory, which excludes the Oman fragments (Table S3b gives B0 with them); Emsian cutoff. Units: guaranteed/possible; Oldest: oldest possible age. History abbreviations: Const., constant; PBDB, proportional to PBDB land and marginal-marine collection density; PBDB −plant, the same with plant-matched collections removed; Decl. 25 and Decl. 10, exponential decline into the past on 25 and 10 Myr timescales; −plant × decl. 10, their product. "None": no bound within the 469.4 Ma limit of the declared domain.

Cutoff (Ma)	Units	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
Emsian (393.47)	48	428.84	428.88	428.45	431.63	none	none
Pragian (410.62)	25	428.75	429.71	429.43	429.75	433.80	451.56
Lochkovian (413.02)	18	429.18	430.09	429.81	430.24	434.25	447.52
Pridoli (419.62)	13	428.53	428.54	428.35	428.92	429.80	429.55

Table S4. Crown vascular-plant bounds (Ma) by younger cutoff, main reading. Abbreviations as in Table S3.

B2 unit or source variant	Units	Constant	PBDB density	Decline 25 Myr
Individual collections/observations	78	428.00	428.03	429.58
Locality–age proxies (primary)	48	428.84	428.88	431.63
Whole localities	47	428.89	428.93	431.75
Region–stage cells	20	432.07	431.91	441.60
Wide-age records restored	73	428.09	428.12	429.79
Uncertain material restored	54	428.60	428.63	431.01
Low-confidence names omitted	46	428.94	428.98	431.87
Weak botanical source omitted	48	428.84	428.88	431.63
Kosov/Podolia links collapsed	47	428.89	428.93	431.75
Curated oldest-record set only	15	434.05	434.62	450.50

Table S5. Crown vascular-plant bounds under alternative units and source choices (main reading, Emsian cutoff). Region–stage cells use midpoint stage and are a deliberately coarse dependence diagnostic.

Region left out	Units	Oldest possible	Constant	PBDB density
Gondwana Australia Antarctica	44	425.0	427.22	427.33
Gondwana South America	47	426.7	428.89	428.87
Tarim	47	426.7	428.89	428.92
peri Gondwana Europe	47	426.7	428.89	428.85
Baltica	45	426.7	428.99	431.80
Kazakhstania	45	426.7	428.99	429.03
Avalonia	44	426.7	429.04	428.95
South China	43	426.7	429.10	429.06
multiple	42	426.7	429.16	428.74
Laurentia	28	426.7	430.45	429.58

Table S6. Leave-one-region-out bounds for crown vascular-plant members, with PBDB exposure recomputed after the same omission. Regions are approximate modern country/state assignments.

Category	History	Nominal p, midpoint ages	Independent-uniform ages, 5–95%	Shared-quantile ages, 5–95%
Land-plant body, without Oman fragments	Constant	0.0066	0.0041–0.062	1.1e-09–0.0062
Land-plant body, without Oman fragments	PBDB collection density	0.02	0.0075–0.088	0.00033–0.022
Land-plant body, without Oman fragments	PBDB density, plant matches removed	0.0019	0.00054–0.039	4.4e-05–0.0047
Land-plant body, without Oman fragments	Decline into past, 25 Myr	1.2e-07	6.2e-09–2.7e-06	2e-21–1.9e-06
Land-plant body, without Oman fragments	Decline into past, 10 Myr	2.5e-20	2.3e-22–4e-17	1.4e-40–8.4e-17
Land-plant body, without Oman fragments	Plant-excluded PBDB × 10 Myr decline	3.9e-09	2e-10–5.4e-08	1.7e-16–1.4e-08
Polysporangiophyte body	Constant	0.0053	0.0046–0.068	6.2e-09–0.0056
Polysporangiophyte body	PBDB collection density	0.016	0.011–0.13	0.00061–0.031
Polysporangiophyte body	PBDB density, plant matches removed	0.0016	0.00086–0.032	4.1e-05–0.0038
Polysporangiophyte body	Decline into past, 25 Myr	4.5e-07	3.1e-08–1.8e-05	7.6e-20–1.1e-05
Polysporangiophyte body	Decline into past, 10 Myr	7.8e-19	1.9e-20–1.8e-15	1.1e-37–4.3e-15
Polysporangiophyte body	Plant-excluded PBDB × 10 Myr decline	8.5e-08	1e-08–1.6e-06	2.9e-15–5.6e-07
Crown vascular-plant body	Constant	0.0085	0.073–0.76	3.1e-09–0.016
Crown vascular-plant body	PBDB collection density	0.0092	0.014–0.27	2.7e-05–0.058
Crown vascular-plant body	PBDB density, plant matches removed	8.6e-05	0.00019–0.016	2.2e-06–0.019
Crown vascular-plant body	Decline into past, 25 Myr	4.9e-06	7e-05–0.016	4.6e-16–0.0019
Crown vascular-plant body	Decline into past, 10 Myr	3.1e-13	2.2e-11–8.4e-07	3.5e-26–8e-07
Crown vascular-plant body	Plant-excluded PBDB × 10 Myr decline	1.5e-05	0.0011–0.044	5.9e-11–0.0064
Land-plant body, with Oman fragments	Constant	5.5e-12	5.2e-13–5.2e-09	—
Land-plant body, with Oman fragments	PBDB collection density	7.5e-08	1.2e-08–4.7e-06	—
Land-plant body, with Oman fragments	PBDB density, plant matches removed	1.1e-10	1.3e-11–2e-08	—
Land-plant body, with Oman fragments	Decline into past, 25 Myr	6.3e-06	3.2e-06–0.00085	—
Land-plant body, with Oman fragments	Decline into past, 10 Myr	7.4e-20	9.6e-22–1.4e-16	—
Land-plant body, with Oman fragments	Plant-excluded PBDB × 10 Myr decline	1.2e-08	1.2e-09–1.5e-07	—

Table S7. Within-record shape diagnostics (conditional-on-maximum Kolmogorov–Smirnov), main reading, under each declared age treatment. Ranges are the 5th–95th percentiles of nominal p across 256 age draws, not minimum-to-maximum ranges.

Category	History	Allocation	5%	Median	95%	Unbounded fraction
Land-plant body, without Oman fragments	Constant	independent uniform	432.25	433.34	434.42	0.00
Land-plant body, without Oman fragments	PBDB collection density	independent uniform	432.02	433.42	436.18	0.00
Land-plant body, without Oman fragments	Decline into past, 25 Myr	independent uniform	434.48	435.76	437.01	0.00
Land-plant body, without Oman fragments	Constant	shared quantile	432.23	433.27	434.35	0.00
Land-plant body, without Oman fragments	PBDB collection density	shared quantile	432.00	433.27	435.99	0.00
Land-plant body, without Oman fragments	Decline into past, 25 Myr	shared quantile	434.48	435.66	436.90	0.00
Polysporangiophyte body	Constant	independent uniform	432.32	433.33	434.43	0.00
Polysporangiophyte body	PBDB collection density	independent uniform	432.07	433.35	436.15	0.00
Polysporangiophyte body	Decline into past, 25 Myr	independent uniform	434.72	435.93	437.22	0.00
Polysporangiophyte body	Constant	shared quantile	432.34	433.38	434.46	0.00
Polysporangiophyte body	PBDB collection density	shared quantile	432.09	433.50	436.23	0.00
Polysporangiophyte body	Decline into past, 25 Myr	shared quantile	434.78	435.97	437.21	0.00
Crown vascular-plant body	Constant	independent uniform	424.31	426.33	428.46	0.00
Crown vascular-plant body	PBDB collection density	independent uniform	425.14	426.71	428.51	0.00
Crown vascular-plant body	Decline into past, 25 Myr	independent uniform	426.19	428.57	431.05	0.00
Crown vascular-plant body	Constant	shared quantile	421.58	424.10	427.03	0.00
Crown vascular-plant body	PBDB collection density	shared quantile	421.22	424.81	427.18	0.00
Crown vascular-plant body	Decline into past, 25 Myr	shared quantile	423.10	425.92	429.34	0.00

Table S8. Age-allocation sensitivity (Emsian cutoff, main reading). Quantiles across allocations are descriptive, not confidence bounds.

S5.4 Recovery before the oldest record

After seeing that steep smooth declines fit poorly, we evaluated a complete stage-break grid: flat and PBDB baselines; breaks at 419.62, 426.7 and 432.9 Ma; older multipliers 1, 0.5, 0.1 and 0.01; all categories and both readings (144 cells). The 426.7 Ma break coincides with the oldest possible main B2 record, so no main B2 unit's age support is changed and the within-record shape statistic is unchanged. The grid is a retrospective sensitivity; the 0.05 rule supplies no nominal, simultaneous or selection-adjusted confidence for it. The 432.9 Ma break is a null control when the bound lies younger. No member was selected to match a clock (Table S9).

Baseline	Break (Ma)	Older multiplier	Changes age support?	Older bound (Ma)	Midpoint shape p
constant	419.62	1	True	428.84	0.00848

Baseline	Break (Ma)	Older multiplier	Changes age support?	Older bound (Ma)	Midpoint shape p
constant	419.62	0.5	True	430.52	0.00424
constant	419.62	0.1	True	444.00	0.00223
constant	419.62	0.01	True	none	0.00191
constant	426.7	1	False	428.84	0.00848
constant	426.7	0.5	False	430.98	0.00848
constant	426.7	0.1	False	448.10	0.00848
constant	426.7	0.01	False	none	0.00848
constant	432.9	1	False	428.84	0.00848
constant	432.9	0.5	False	428.84	0.00848
constant	432.9	0.1	False	428.84	0.00848
constant	432.9	0.01	False	428.84	0.00848
PBDB density	419.62	1	True	428.88	0.00918
PBDB density	419.62	0.5	True	430.60	0.0186
PBDB density	419.62	0.1	True	451.09	0.00729
PBDB density	419.62	0.01	True	none	0.00576
PBDB density	426.7	1	False	428.88	0.00918
PBDB density	426.7	0.5	False	430.98	0.00918
PBDB density	426.7	0.1	False	458.47	0.00918
PBDB density	426.7	0.01	False	none	0.00918
PBDB density	432.9	1	False	428.88	0.00918
PBDB density	432.9	0.5	False	428.88	0.00918
PBDB density	432.9	0.1	False	428.88	0.00918
PBDB density	432.9	0.01	False	428.88	0.00918

Table S9. Retrospective stage-break grid for crown vascular-plant members, main reading (`models/f3/loop3/out/suppres`

S5.5 Biological delay

Fixed delays of 0, 5, 10 and 20 Myr map the recoverability bound to a lineage bound and shift the lineage domain by the same amount. A delay is an assumed mapping, not inferred from this record or transferred from the spore comparison (Table S10).

Category	Assumed delay (Myr)	Recoverability bound	Lineage bound
Land-plant body, without Oman fragments	0	434.51	434.51
Land-plant body, without Oman fragments	5	434.51	439.51
Land-plant body, without Oman fragments	10	434.51	444.51
Land-plant body, without Oman fragments	20	434.51	454.51
Polysporangiophyte body	0	434.62	434.62
Polysporangiophyte body	5	434.62	439.62
Polysporangiophyte body	10	434.62	444.62

Category	Assumed delay (Myr)	Recoverability bound	Lineage bound
Polysporangiophyte body	20	434.62	454.62
Crown vascular-plant body	0	428.84	428.84
Crown vascular-plant body	5	428.84	433.84
Crown vascular-plant body	10	428.84	438.84
Crown vascular-plant body	20	428.84	448.84

Table S10. Fixed-delay mapping, constant recovery, main reading.

S5.6 Simulation checks

Each of 144 cells has 3,000 replicates (seed 20261101) at the reviewed unit counts and recoverability endpoints of 432, 445 and 458.2 Ma. Each correctly specified history is checked with exact and stage-censored dates. The checks then mismatch constant and steeply declining histories, repeat underlying units two or four times, and impose 10 or 20 Myr delays with and without the correct adjustment. Coverage counts no-bound outcomes as covering. Claim frequency and coverage among bounded claims are reported separately, because 95% unconditional coverage does not imply 95% coverage after selecting bounded outcomes. These runs verify the declared procedures and expose their failure modes. They do not qualify the fossils (Table S11; all categories in `simulation_results.csv`).

Scenario	True	Fitted	Ages	Dup.	True delay	Adj.	Claims	Coverage (95% CI)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.955 (0.947–0.962)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	1.000 (0.998–1.000)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.949 (0.941–0.957)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	0.985 (0.980–0.989)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.945 (0.936–0.953)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	0.983 (0.977–0.987)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	1.000	0.945 (0.936–0.952)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	1.000	0.956 (0.948–0.963)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	1.000	0.958 (0.951–0.965)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	1.000	0.990 (0.985–0.993)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	0.212	0.941 (0.932–0.949)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	0.070	0.966 (0.959–0.972)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.139	0.950 (0.942–0.958)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.078	0.998 (0.996–0.999)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.065	0.949 (0.940–0.956)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.034	0.966 (0.959–0.972)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.053	0.950 (0.941–0.957)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.032	0.968 (0.961–0.974)

Scenario	True	Fitted	Ages	Dup.	True delay	Adj.	Claims	Coverage (95% CI)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.950 (0.942–0.958)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.999 (0.998–1.000)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.957 (0.949–0.963)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.967 (0.960–0.973)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.953 (0.945–0.960)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.999 (0.998–1.000)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.954 (0.946–0.961)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	1.000 (0.998–1.000)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.948 (0.939–0.955)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	0.983 (0.977–0.987)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.952 (0.944–0.959)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	0.954 (0.946–0.961)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.175	0.951 (0.943–0.958)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.119	0.992 (0.988–0.995)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.069	0.954 (0.946–0.961)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.049	0.951 (0.942–0.958)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.045	0.956 (0.948–0.962)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.040	0.960 (0.953–0.967)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.241 (0.226–0.257)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.104 (0.094–0.116)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.034 (0.028–0.041)
duplicated horizons	Const.	Const.	exact	2	0	0.0	1.000	0.782 (0.767–0.797)
duplicated horizons	Const.	Const.	exact	4	0	0.0	1.000	0.529 (0.511–0.546)
recoverability delay unadjusted	Const.	Const.	exact	1	10	0.0	1.000	0.000 (0.000–0.001)
recoverability delay adjusted	Const.	Const.	exact	1	10	10.0	1.000	0.947 (0.938–0.954)
recoverability delay unadjusted	Const.	Const.	exact	1	20	0.0	1.000	0.000 (0.000–0.001)

Scenario	True	Fitted	Ages	Dup.	True delay	Adj.	Claims	Coverage (95% CI)
recoverability delay adjusted	Const.	Const.	exact	1	20	20.0	1.000	0.952 (0.943–0.959)

Table S11. Simulation results for crown vascular-plant counts. True and Fitted: generating and fitted recording histories (abbreviations as in Table S3); Dup.: duplication factor; True delay: imposed delay (Myr) between lineage origin and the start of recording; Adj.: delay adjustment applied in the fit; Claims: fraction of replicates giving a bound; Coverage: of the lineage or recording endpoint as targeted, with Wilson 95% interval.

S5.7 Figures from the reviewed analysis

S6. From specimens to nodes

The specimen-to-node pilot separates specimen observations, accepted event readings, encounter, trait persistence and node interpretation (`data/pilot/`, `docs/pilot-traits.md`). We reused its source-adjudicated combinations, Hooper et al.'s deposited crown-state outputs and the checked stochastic-map analyses without rerunning them. Aperture (X21) and dispersal unit (X26) were modelled separately, so their marginal supports do not define a joint ancestral phenotype. The X26 monad category combines trilete and inaperturate monads; it is not an automatic absence of cryptospory.

For M2, a proximal trilete aperture is modal at the embryophyte crown across the deposited methods and trees. X26 has a dyad-dominated alternative while X21 stays trilete-dominated, so no joint phenotype probability follows. On the converged extant-tree runs, 77 of 3,382 trilete starts depart along the path to crown Tracheophyta: conditional persistence 0.977 (Monte Carlo SE 0.0026). These trees put crown Embryophyta at 508 Ma, not at Hooper et al.'s 457–450 Ma. Fixed-rate retention scenarios to 452.8 Ma give an extant-tree median of 0.989 (95% interval 0.895–0.995) from 459.1 Ma and 0.894 (0.332–0.956) from 515.5 Ma. These are model-conditional aperture retention scenarios, not observed survival, and they are not multiplied into any coverage.

For M3, the *Cooksonia barrandei* holotype links trilete production and multiple sporangia directly, but its tracheids are unknown. The Irish Homerian material has a conditional sporangial interpretation and no direct demonstration of the anatomy known from younger *C. pertoni*. A total-group or stem occurrence and a crown assignment remain distinct. For M1, cryptospore persistence has not been assessed.

S7. Analyses superseded during review

Analyses and wording were replaced after independent review. The record is kept so that readers can see what changed and why.

The Loop 3 body-record pilot (`docs/loop3-pilot.md`, provisional, by Claude) reported logistic-ramp limits of 452.6, 454.1 and 438.0 Ma for B0, B1 and B2. It read them as favouring the young clock, with a 15–25 Myr preservation lag. Codex's review found:

- stage collection totals passed where per-Myr density was required, counting stage duration twice. The unit correction alone moved B0's logistic limit from 452.57 to 436.78 Ma and flipped the selected capacity between grid boundaries;
- medians of imputed-age limits presented as 95% limits;
- false replacement of distinct PBDB localities through country, genus and age overlap;
- collection numbers treated as independent beds;
- a category gate that let names judged implausible into B1 and B2;
- two different papers sharing the source identifier `wang2026`.

All were repaired. The fixed-history endpoint calculation (S5.2) replaced the profiled ramp. Its numbers and interpretation govern; the pilot's do not (`docs/audits/2026-11-01-loop3-methods-review.md`, `docs/loop3-results.md`).

The GPT-6 Pro review (`docs/audits/2026-11-07-gpt6pro-triage.md`). An independent review of commit `60c406b` found that the earlier drafts described the 515.5 Ma maximum as resting only on affinity, omitting the published preservation-control argument; kept the Oman fragments out of the primary land-plant inventory; described the recording rate as fossilisation alone; did not show that 18 of the 28 older Murzuq zeros are younger than 459.1 Ma; and read shape diagnostics on allocated ages as rejections. All

are corrected here; analyses that the review requested (interval feasibility, an age–intensity frontier, shared timescale-boundary scenarios, Murzuq block dependence and a public raw-input rebuild) are in hand.

Review C wording (`docs/audits/2026-11-05-review-c.md`, `docs/audits/2026-11-05-review-c-methods.md`). The previous draft called the B0–spore separation a demonstrated, observed lag; applied the $\rho f \approx 0.09$ product to both Murzuq history classes when it holds for the constant class only; reported the QSIM minimum ratios as percentages of all spores (0.307 as 31% rather than a 23.5% share); showed shape diagnostics for one age treatment only; and presented the tenfold suppression and the 15–23 Myr delays as if they were estimated. All are corrected here, and the Oman sensitivity was added.

Loop 2 corrections (`docs/loop2-close.md`, `docs/audits/2026-11-01-loop2-close-claude-response.md`) fixed a misread H0c column, an overgeneralised exponent sensitivity, an unsupported circularity claim and an overstated examination count in an earlier draft.

S8. Reproduction and navigation

From the repository root:

```
uv run --with numpy --with pandas --with scipy --with pytest python -m pytest models/f3/loop3 -q
uv run manuscript/figures.py
python3 manuscript/tables.py
uv run tools/refs.py build
make -C manuscript
python3 manuscript/package.py
```

The body-record analysis is reproduced by `models/f3/loop3/build_reviewed.py`, `analysis.py`, `simulation.py`, `suppression_sensitivity.py` and `figures_reviewed.py`. Hash receipts bind their inputs and outputs. Frozen Loop 1 and Loop 2 runs are retained without regeneration. Third-party deposits are fetched by `data/external/scripts/deposits_fetch.py` and verified against `MANIFEST.tsv`. `manuscript/claims.csv` links each main claim to its evidence and limits.

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Supplementary Information

Fig. S1.

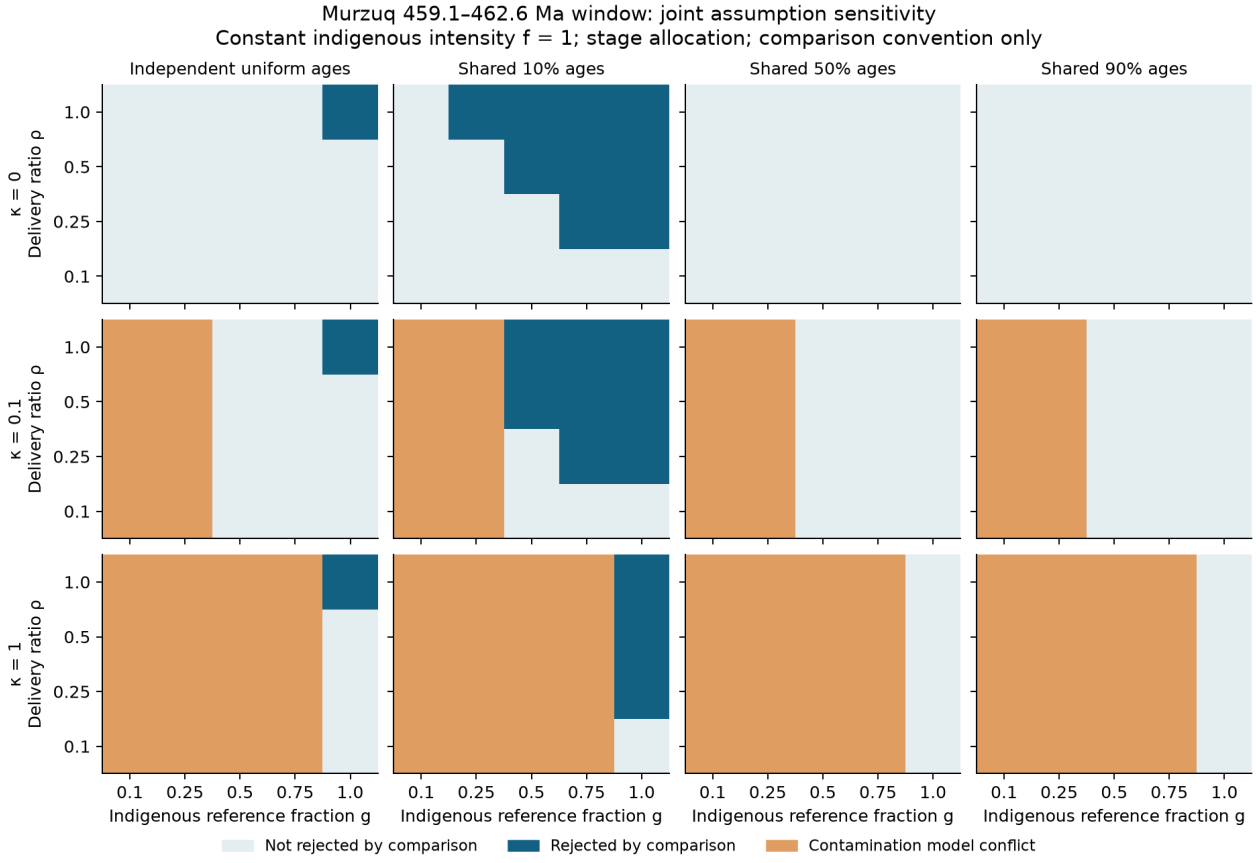


Figure 1: Joint sensitivity of the short Murzuq window. Colours report an uncalibrated comparison or model-conflict flag, not confidence or probabilities over geology.

Fig. S2.

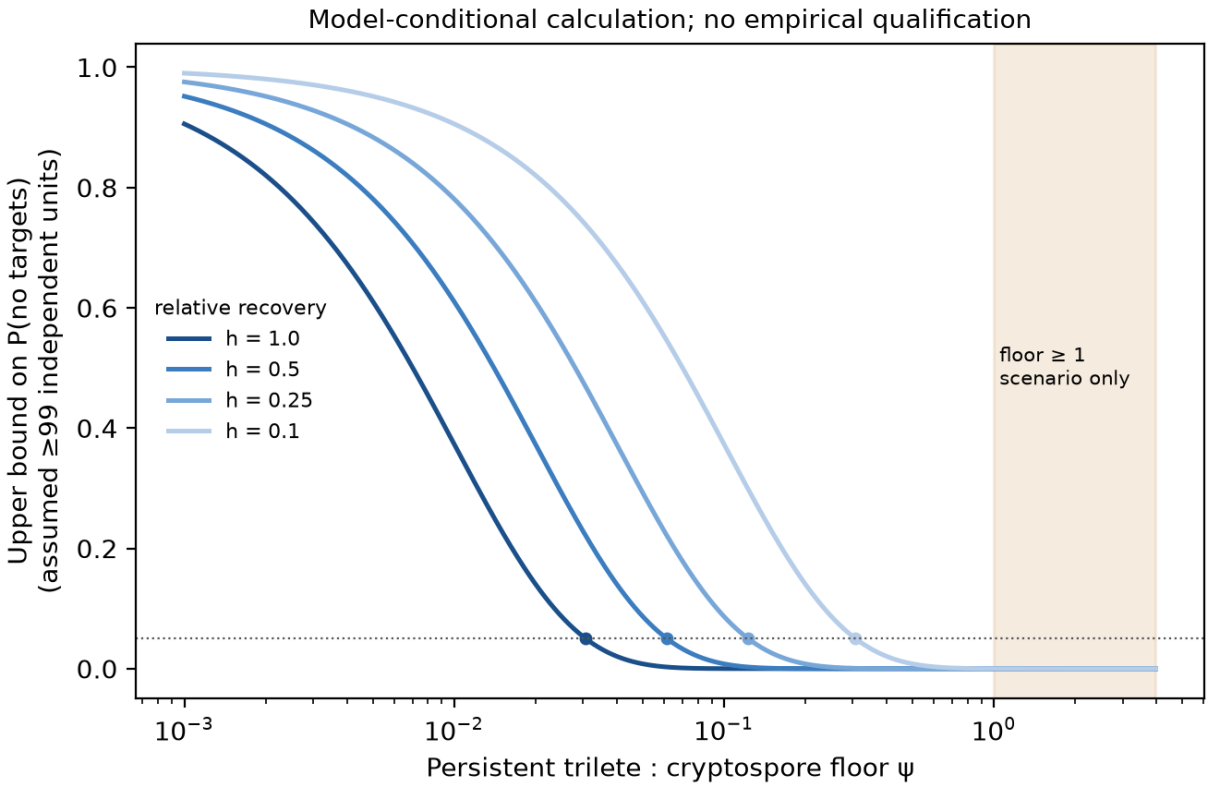


Figure 2: Upper bound on the all-zero probability under the hypothetical independent, co-ascertained unit model; points mark the 0.05 threshold.

Fig. S3.

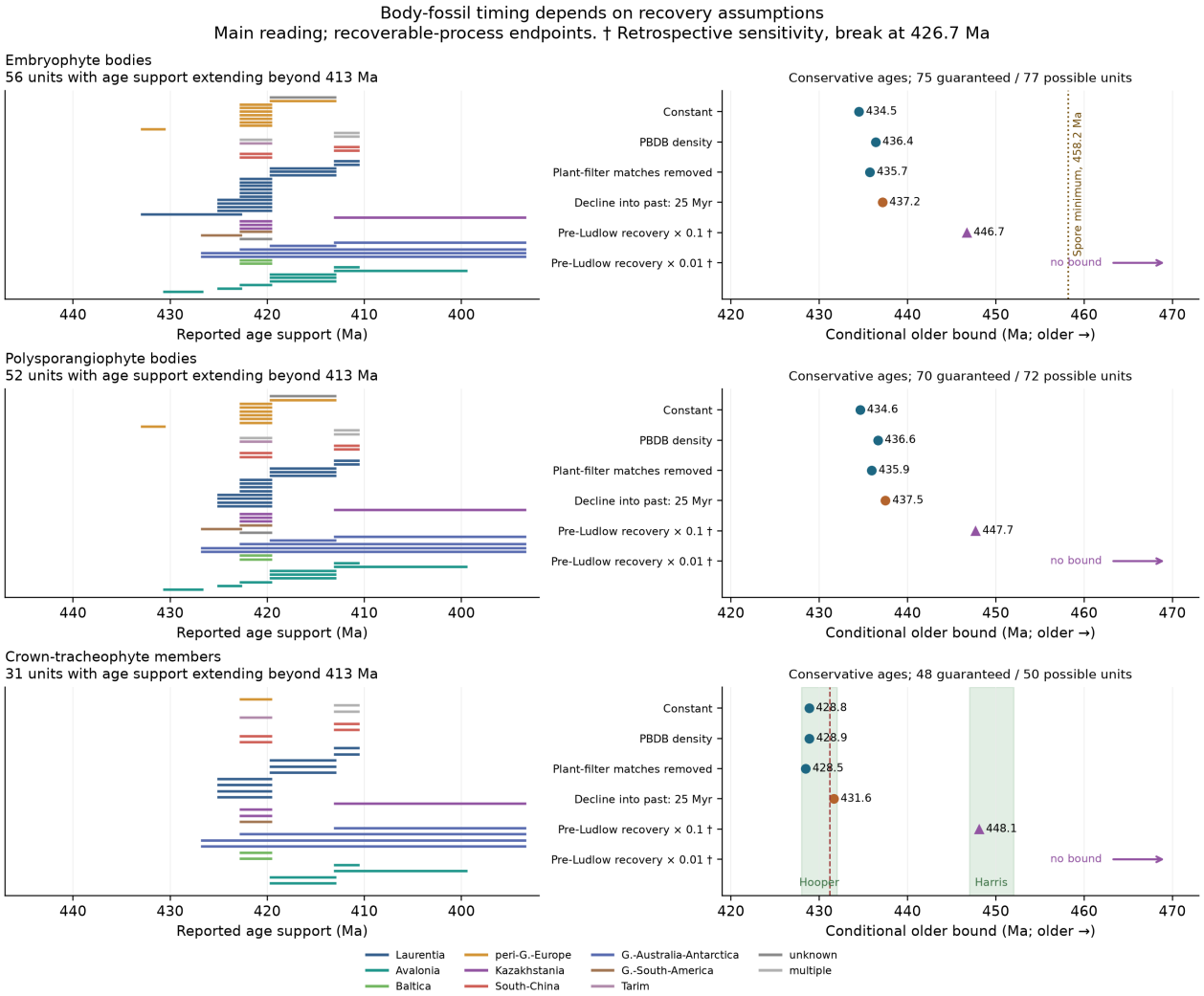


Figure 3: Reviewed main macrofossil figure as delivered by the analysis (`models/f3/loop3/out/fig_macro_recovery.pdf`).

Fig. S4.

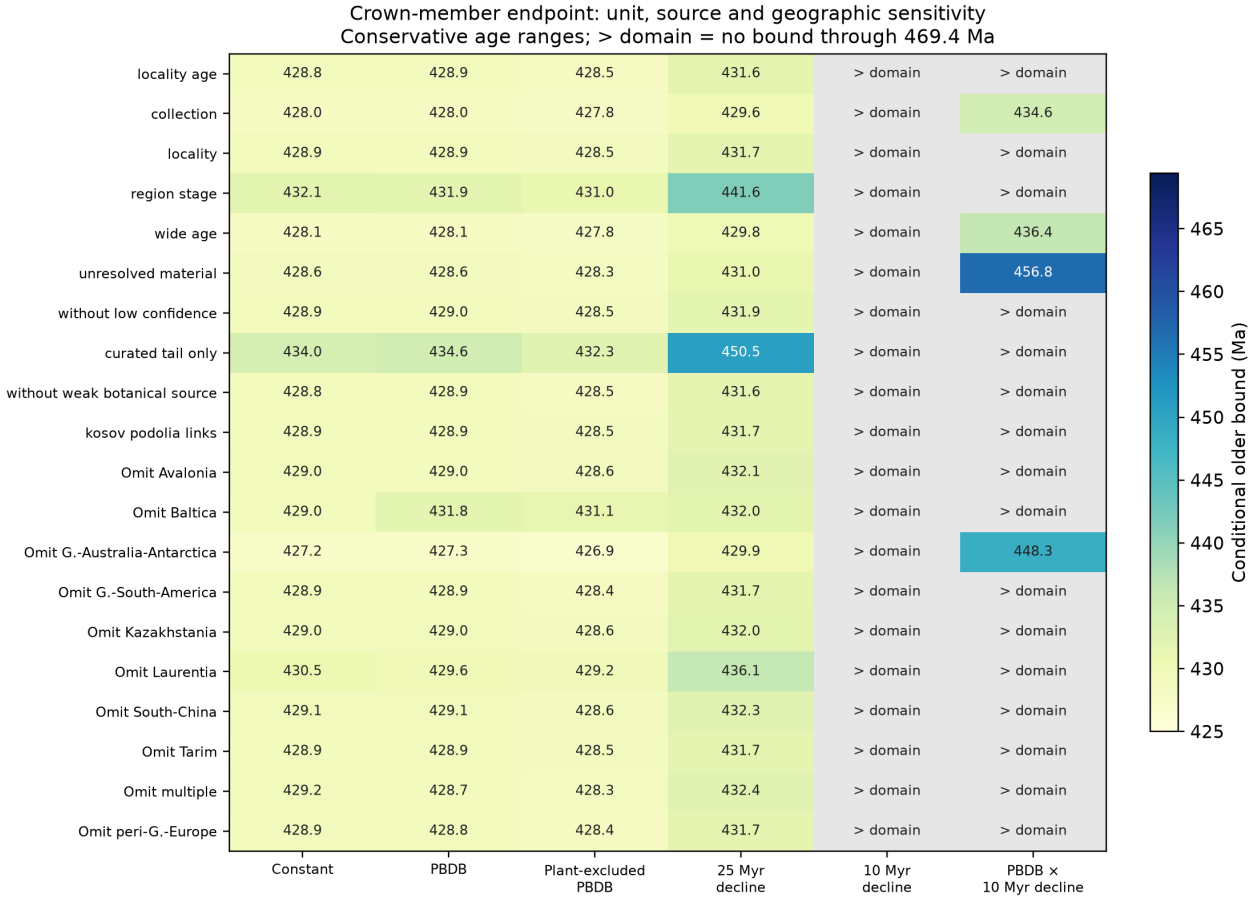


Figure 4: Sensitivity of the crown vascular-plant bound to units, sources, regions and age rules.

Fig. S5.

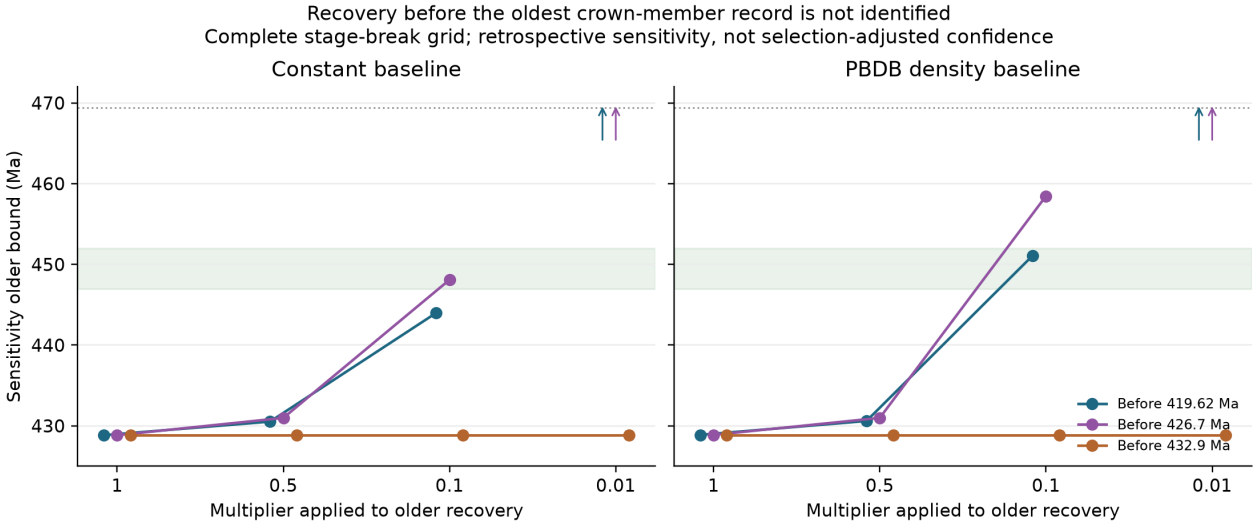


Figure 5: Retrospective recovery-suppression grid.